

## ASCII MASTER FLOW – PANEL 1/12: Urban risk system

HAZARD + DENSITY + EXPOSED ASSETS + UNEQUAL VULNERABILITY

-> INTERDEPENDENT WATER POWER TRANSPORT HEALTH TELECOM

-> CASCADING SERVICE FAILURE

-> SYSTEMIC URBAN DISASTER

## ASCII MASTER FLOW – PANEL 2/12: Criticality test

ASSET / NETWORK / SERVICE

IF FAILURE -> DISPROPORTIONATE LIFE / HEALTH / GOVERNANCE EFFECT

THEN CRITICAL / LIFELINE FUNCTION

PRIORITY DEPENDS ON CONSEQUENCE, NOT OWNERSHIP OR SIZE

## ASCII MASTER FLOW – PANEL 3/12: Interdependency web

POWER -> WATER PUMPS / HOSPITALS / TELECOM

TELECOM -> WARNING / DISPATCH / PAYMENT

TRANSPORT -> STAFF / SUPPLY / REPAIR ACCESS

SHARED NODE FAILURE -> MULTI-SYSTEM CASCADE

## ASCII MASTER FLOW – PANEL 4/12: Resilience triad

ROBUSTNESS -> RESIST

REDUNDANCY / DIVERSITY -> ALTERNATE CAPACITY

RAPID RECOVERY -> RESTORE PRIORITY FUNCTION

COMPLETE RESILIENCE NEEDS ALL THREE

## ASCII MASTER FLOW – PANEL 5/12: Continuity plan

DEFINE MINIMUM SERVICE + PRIORITY USERS  
MAP STAFF SUPPLY DATA SITE AND UTILITY DEPENDENCIES  
PROVIDE BACKUP / ALTERNATE / MANUAL MODE  
SET RESTORATION ORDER TIME AND COMMUNICATION

## ASCII MASTER FLOW – PANEL 6/12: Urban planning stack

LAND USE / FLOODPLAIN / DRAINAGE PATH  
BUILDING AND DEVELOPMENT CONTROL  
WATER BODY DRAIN SOLID-WASTE AND MAINTENANCE  
WARD-CITY-CATCHMENT COORDINATION

## ASCII MASTER FLOW – PANEL 7/12: Resilience audit

- 1 CRITICAL FUNCTION AND USERS
- 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS
- 3 BACKUP DURATION + RECOVERY TIME
- 4 PRIORITISED RETROFIT CONTINUITY AND DRILL ACTION

## ASCII MASTER FLOW – PANEL 8/12: Essential facility shell

STRUCTURE / FIRE / FLOOD SAFETY  
POWER WATER CLINICAL SUPPLY AND TELECOM  
ACCESS STAFF REFERRAL AND SECURITY  
CONTINUE CARE WHILE DAMAGE IS REPAIRED



## ASCII MASTER FLOW – PANEL 9/12: Equity access test

WHO LIVES IN HIGHEST EXPOSURE?

WHO DEPENDS ON INTERRUPTIBLE OR INFORMAL SERVICES?

WHO CAN REACH WARNING SHELTER HEALTH AND TRANSPORT?

RISK REDUCTION WITHOUT EXCLUSIONARY DISPLACEMENT

## ASCII MASTER FLOW – PANEL 10/12: Governance map

ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT

SECTOR REGULATOR + SDMA / NDMA

COORDINATION DOES NOT ERASE OWNER DUTY

RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION

MUST REMEMBER: Urban and critical-infrastructure resilience maps interdependent power,...

## ASCII MASTER FLOW – PANEL 11/12: Smart resilience firewall

SENSORS / DIGITAL TWIN / COMMAND PLATFORM -> AWARENESS

PRIVACY + CYBERSECURITY + INTEROPERABILITY

OFFLINE FALLBACK + HUMAN ACCOUNTABILITY

EFFICIENCY WITHOUT BUFFER -> BRITTLE SYSTEM

CLOSE DISTINCTION: Critical is not merely expensive; redundancy is not unused...

## ASCII MASTER FLOW – PANEL 12/12: Urban resilience answer spine

DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES

MAP CASCADES -> ROBUSTNESS REDUNDANCY CONTINUITY

ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE

USE CDRI / TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name service level, dependency,...